THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The sum of coefficients in the balanced equation $Fe + H_2O \longrightarrow Fe_3O_4 + H_2$ is: 1

(a) 12 **(b)** 8 **(c)** 9 **(d)** 10

2. The pH of a solution is 5. If its pH becomes 3, its H^+ concentration: 1

(a) decreases 100 times **(b)** increases 2 times
(c) decreases 2 times **(d)** increases 100 times

3. Which of the following will not react with dilute HCl? 1

(a) iron **(b)** zinc **(c)** magnesium **(d)** silver

4. The number of covalent bonds in butane C_4H_{10} is: 1

(a) 10 **(b)** 13 **(c)** 14 **(d)** 12

5. In photosynthesis, the oxygen released comes from: 1

(a) chlorophyll **(b)** water **(c)** glucose **(d)** carbon dioxide

6. In a reflex arc, the correct order is: 1

(a) receptor -> motor neuron -> spinal cord -> sensory neuron -> effector
(b) receptor -> sensory neuron -> spinal cord -> motor neuron -> effector
(c) effector -> motor neuron -> spinal cord -> sensory neuron -> receptor
(d) receptor -> spinal cord -> brain -> effector

7. In which of the following does the offspring differ least from the parent? 1

(a) Offspring produced by cross-pollination **(b)** Offspring produced by seeds
(c) Offspring produced by budding in Hydra **(d)** Offspring produced by fertilisation

8. A heterozygous tall plant (Tt) is crossed with a dwarf plant (tt). The fraction of dwarf plants in the offspring is 1

(a) 3/4 **(b)** 1/2 **(c)** 0 **(d)** 1/4

9. An object is placed at the centre of curvature of a concave mirror of focal length 12 cm. The distance of the image from the mirror is 1

(a) 24 cm **(b)** 12 cm **(c)** 6 cm **(d)** 36 cm

10. A person with myopia has a far point at 100 cm. The lens needed has a power of 1

(a) -0.1 D **(b)** -1 D **(c)** +1 D **(d)** +10 D

11. The number of electrons passing per second through a conductor carrying 3.2 A is 1

(a) 3.2×10^{19} **(b)** 2×10^{19} **(c)** 2×10^{18} **(d)** 5×10^{18}

12. A compass needle placed below a wire carrying current from south to north is deflected. If the current is reversed, the deflection 1

(a) stays the same **(b)** becomes zero **(c)** doubles **(d)** is reversed

13. In the food chain Phytoplankton, Zooplankton, Small fish, Large fish the secondary consumer is 1

(a) zooplankton **(b)** phytoplankton **(c)** small fish **(d)** large fish

14. In $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$, the substance reduced is: 1

(a) Fe_2O_3 **(b)** Al **(c)** Fe **(d)** Al_2O_3

15. The solid residue obtained on heating washing soda crystals is: 1

(a) sodium hydroxide **(b)** sodium chloride
(c) anhydrous sodium carbonate **(d)** sodium bicarbonate

16. Which of the following is correct about ionic compounds? 1

(a) They are insoluble in water **(b)** They are soft
(c) They are gaseous **(d)** They have high melting points

17. CH_3CH_2OH on heating with excess conc. H_2SO_4 at 443 K gives: 1

(a) CH_3CHO **(b)** $CH_2{=}CH_2$ **(c)** CH_3CH_3 **(d)** CH_3COOH

18. Which has the longest small intestine? 1

(a) carnivores like tiger **(b)** humans
(c) birds **(d)** herbivores such as the cow

19. Why are shoots positively phototropic and roots negatively phototropic? 1

(a) roots need light
(b) roots have no auxin
(c) shoots need gravity
(d) auxin affects stem and root cells in opposite ways

20. The thread-like structure on which anther is held is the 1

(a) style **(b)** ovule **(c)** stigma **(d)** filament

21. The sex of a baby is determined at the time of 1

(a) puberty **(b)** fertilisation **(c)** birth **(d)** implantation

22. If n = sin i / sin r for glass is 1.5 and sin i = 0.75, then sin r is 1

(a) 0.75 **(b)** 0.5 **(c)** 0.25 **(d)** 1.125

23. A beam of white light is passed through a prism and a screen shows 7 colours. Replacing white by red light gives 1

(a) 7 colours **(b)** white light
(c) no light **(d)** a single red patch

24. Three resistors of 6 ohm each are connected in parallel. The equivalent resistance is 1

(a) 6 ohm **(b)** 2 ohm **(c)** 3 ohm **(d)** 18 ohm

25. A straight conductor carrying current from east to west is placed in a magnetic field pointing down. The force is directed 1

(a) to the south **(b)** downward **(c)** to the north **(d)** upward

26. In a food chain, DDT concentration is highest in 1

(a) zooplankton **(b)** eagle **(c)** phytoplankton **(d)** small fish

27. The colour of copper sulphate solution fades when an iron nail is dipped in it because: 1

(a) water is formed **(b)** $FeSO_4$ (pale green) is formed
(c) iron turns blue **(d)** copper sulphate evaporates

28. A student tests an unknown solution with litmus and gets no change. The solution is most likely: 1

(a) basic **(b)** acidic **(c)** neutral **(d)** strongly acidic

29. Iron articles are painted to prevent rusting because paint: 1

(a) makes iron heavy **(b)** makes iron reactive
(c) keeps air and water away **(d)** reacts with iron

30. Which of these is NOT a property of carbon? 1

(a) catenation **(b)** forms multiple bonds
(c) tetravalency **(d)** forms ionic bonds with chlorine

31. The blood vessel carrying deoxygenated blood from the heart to the lungs is: 1

(a) pulmonary artery **(b)** vena cava **(c)** aorta **(d)** pulmonary vein

32. A person has frequent thirst and urination and sugar in urine. Which hormone is deficient? 1

(a) thyroxine **(b)** insulin **(c)** adrenaline **(d)** growth hormone

33. How many chromosomes would a human egg cell have, if the body cell has 46? 1

(a) 23 **(b)** 46 **(c)** 92 **(d)** 12

34. A trait that is controlled by a recessive allele appears in a plant when 1

(a) both alleles are recessive **(b)** no allele is present
(c) both are dominant **(d)** one allele is recessive

35. An object is placed 40 cm from a convex lens of f = 20 cm. The image distance is 1

(a) 60 cm **(b)** 20 cm **(c)** 40 cm **(d)** 10 cm

36. If the atmosphere of the earth were absent, the day sky would appear 1

(a) dark **(b)** red **(c)** blue **(d)** white

37. The power of an electric device that draws 5 A at 220 V is 1

(a) 1100 W **(b)** 225 W **(c)** 1.1 W **(d)** 44 W

38. If a magnet is moved faster into a coil, the induced current 1

(a) decreases **(b)** increases **(c)** remains the same **(d)** becomes zero

39. Segregation of waste means 1

(a) burning
(b) dumping in sea
(c) burying
(d) separating biodegradable from non-biodegradable

40. The coefficient of O_2 in the balanced equation $C_3H_8 + O_2 \longrightarrow CO_2 + H_2O$ is: 1

(a) 7 **(b)** 5 **(c)** 3 **(d)** 4

41. Which of the following will have the highest H^+ ion concentration? 1

(a) pH 7 **(b)** pH 1 **(c)** pH 4 **(d)** pH 10

42. A metal M reacts with cold water giving hydrogen and a base. M could be: 1

(a) copper **(b)** silver **(c)** sodium **(d)** gold

43. The number of covalent bonds in propane C_3H_8 is: 1

(a) 10 **(b)** 11 **(c)** 9 **(d)** 8

44. Which of the following is NOT a requirement for photosynthesis? 1

(a) water **(b)** oxygen **(c)** sunlight **(d)** chlorophyll

45. The connection between sensory and motor neurons inside the spinal cord is made by: 1

(a) relay neuron **(b)** effector **(c)** axon **(d)** dendrite

46. A unicellular organism that divides into many cells inside a protective cyst is 1

(a) Plasmodium **(b)** Yeast **(c)** Paramecium **(d)** Amoeba

47. Two tall plants produce 75 tall and 25 dwarf plants. The parents were most likely 1

(a) both Tt **(b)** both TT **(c)** TT and Tt **(d)** Tt and tt

48. A concave mirror of f = 15 cm forms an image of the same size as the object. The object distance is 1

(a) 7.5 cm **(b)** 15 cm **(c)** 45 cm **(d)** 30 cm

49. A person's near point is 50 cm. To read at 25 cm, he needs a lens of power 1

(a) +2 D **(b)** +0.5 D **(c)** +4 D **(d)** -2 D

50. A charge of 600 C flows in 5 minutes. The current is 1

(a) 120 A **(b)** 0.5 A **(c)** 3 A **(d)** 2 A

END OF QUESTION PAPER **Total: 50 Marks**

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